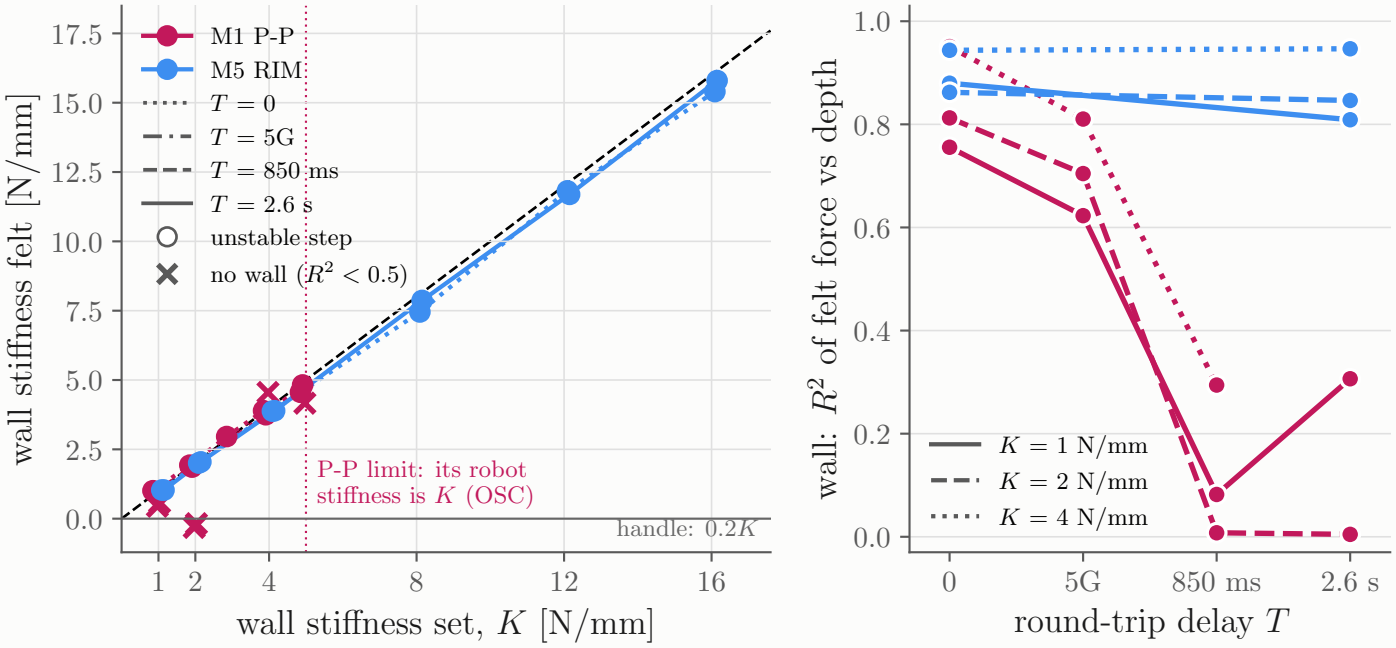

M1 P-P
M5 RIM
$T = 0$
$T = 5$G
$T = 850$ ms
$T = 2.6$ s
unstable step
no wall ($R^2 < 0.5$)
P-P limit: its robot stiffness is K (OSC)
handle: $0.2K$
wall stiffness felt [N/mm]
wall stiffness set, K [N/mm]
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
1
2
4
8
12
16
wall: R^2 of felt force vs depth
round-trip delay T
1.0
0.8
0.6
0.4
0.2
0.0
$K = 1$ N/mm
$K = 2$ N/mm
$K = 4$ N/mm
0
5G
850 ms
2.6 s